Executive Summary

The EMBC architecture model contains the component XeN (Xenon-Nitrogen compound) as a functional modifier with a mass fraction of

$$w_{XeN} = 0.10$$

and an assumed molar mass of

$$M_{XeN} = 145 \text{ g/mol}$$

.

Problem: Stable Xe-N compounds at normal pressure do not exist according to the current state of research. The only experimentally proven Xe-N interactions require pressures above 5 GPa (Howie et al., 2016) or

$$\geq 146 \text{ GPa}$$

for covalent bonds (Peng et al., 2015). XeN is therefore a conceptual placeholder, not a synthesizable material.

Decision: XeN is replaced by a dual-dopant system consisting of F-POSS (Fluorinated Polyhedral Oligomeric Silsesquioxanes) and HfON (Hafnium Oxynitride). This system covers all three identified XeN functions, is fully commercially available, and eliminates Risk R-04 from the EMBC risk register.

Modified Core Formula:

$$Y^*_{Core} = [n' \cdot C_{12}H_{14}O_5Si(F_x) + y \cdot [F\text{-}POSS] + p \cdot [HfON] + m' \cdot (SiR_2O)] + [BNNT] + [GaN]$$

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1. The XeN Problem

1.1 State of Research

Finding	Source
$Xe(N_2)_2$ only forms above 5 GPa as a van der Waals complex (not covalent)	Howie et al., Scientific Reports 6, 34896 (2016)
Covalent XeN theoretically predicted starting at 146 GPa (not experimentally confirmed)	Peng et al., Phys. Rev. B 92, 094104 (2015)
All known stable Xenon compounds at normal pressure contain Fluorine or Oxygen, never Nitrogen	State of Noble Gas Chemistry 2026

Consequence: XeN can neither be procured nor synthesized. It is not a supply chain issue, but a fundamental chemical obstacle.

1.2 Functional Role of XeN in the EMBC

Before a replacement can be selected, its function must be isolated. From the EMBC equation system:

Function	Rank	Justification
Dielectric Tuning	Primary	XeN modifies $$\epsilon_h \rightarrow$$ influences $$\epsilon_{eff} \rightarrow$$ determines $$f_0 = 13.56 \text{ MHz}$$ via $$C_{mat} = \epsilon_0 \epsilon_{eff} \cdot A/d$$
Phonon Damping	Secondary	Xe (131 u) as a heavy atom scatters phonons $$\rightarrow$$ thermal isolation of the resonance layer. Not modeled in the equation system.
N-Coordination	Secondary	Nitrogen acts as a donor for Si-N coordination to the organosilicate matrix. Not modeled in the equation system.

Decision Logic: The replacement strategy must obligatorily cover the primary function (dielectric tuning) and cover the secondary functions if possible.

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2. The Solution: Dual-Dopant System F-POSS + HfON

2.1 Why two components instead of one?

No single candidate from the combined Auron/Lyra analysis (7 materials evaluated) covers all three functions simultaneously without compromises. A dual-dopant system solves this problem and offers a decisive advantage over the original:

Two independent tuning parameters for

$$\epsilon_h$$

: F-POSS lowers

$$\epsilon_h$$

, HfON raises

$$\epsilon_h$$

. By varying the proportions

$$y$$

and

$$p$$

,

$$\epsilon_{eff}$$

can be precisely adjusted to the target value yielding

$$f_0 = 13.56 \text{ MHz}$$

. XeN only offered one tuning parameter with an unknown

$$\epsilon$$

-contribution.

2.2 Component Profiles

Component 1: F-POSS (Fluorinated Polyhedral Oligomeric Silsesquioxanes)
Source of recommendation: Lyra (Candidate #1, highest overall rating 4.4/5)

Property	Value	Relevance for EMBC
Basic Structure	Si_8O_{12} -cage with fluorinated substituents	Identical backbone as organosilicate host $\rightarrow$ seamless integration
Permittivity (ϵ)	2.0–2.5 (tunable via fluorination degree)	ϵ_h -reduction $\rightarrow$ fine-tuning downwards
Thermal Stability	$> 300^\circ C$ (depending on substituents)	Within the EMBC thermal budget ($\leq 450^\circ C$)
Sol-Gel Compatibility	✓ Native (reacts directly into the silicate network)	No additional process step required
Fluorination	Matches the F_x groups in the host $C_{12}H_{14}O_5Si(F_x)$	Chemical coherence with the matrix
Healing-Cycle Compatibility	✓ Does not disrupt SiR_2O siloxane healing	Same chemical family (silicates)
Availability	Commercial (Hybrid Plastics, Sigma-Aldrich, etc.)	No sourcing risk
Molar Mass	$\approx 1000 - 2000\ \mathrm{g/mol}$ (depending on substituents)	Stoichiometric adjustment required

Function in the dual system: Structural matrix integration, ϵ_h -reduction, free volume, chemical compatibility.

Component 2: HfON (Hafnium Oxynitride)

Source of recommendation: Auron (Cross-find from cross-reference analysis, rating 4.3/5)

Property	Value	Relevance for EMBC
Permittivity (ϵ)	14–21 (tunable via O:N ratio)	ϵ_h -increase $\rightarrow$ fine-tuning upwards
Atomic Mass (Hf)	178.5 u (heavier than $Xe = 131\ \mathrm{u}$)	Phonon damping (Secondary function)
Nitrogen in Lattice	✓ N directly in the crystal structure	N-incorporation (Secondary function)
Thermal Stability	$> 2000^\circ C$	Well above EMBC thermal budget
Crystallization Temp.	$\approx 300^\circ C$ higher than HfO_2	Stably amorphous across the process window
CMOS Compatibility	✓ Industry standard (High-k gate dielectric)	Foundry experience, ALD recipes available
Availability	Commercial (ALD precursors at STREM, Sigma-Aldrich, etc.)	No sourcing risk
Integration Form	Nanoparticle dispersion in matrix	Requires dispersion protocol

Function in the dual system: Dielectric increase, heavy atom effect, nitrogen provision.

2.3 Function Coverage Comparison

XeN Function	XeN (Original)	F-POSS alone	HfON alone	F-POSS + HfON
Dielectric Tuning ($\uparrow\downarrow$)	unknown	$\downarrow$ only reduction	$\uparrow$ only increase	**bidirectional**
Heavy Atom (Phonons)	$Xe = 131\ \mathrm{u}$	❌	$Hf = 178.5\ \mathrm{u}$	✓
Nitrogen in Lattice	✓ (theoretical)	❌	✓	✓
Matrix Integration	unknown	✓ native	Nanoparticles	✓
Commercially Available	❌ non-existent	✓	✓	✓
Sol-Gel Compatible	❌ unknown	✓	✓	✓
Healing-Cycle Neutral	❌ unknown	✓	✓	✓

Result: The dual system is superior to the original in every measurable dimension, because it uses real materials with known properties instead of a conceptual placeholder.

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3. Modified Core Formula

3.1 New YCore Formula

Original:

$$Y_{Core} = [n \cdot C_{12}H_{14}O_5Si(F_x) + 2 \cdot XeN + m \cdot (SiR_2O)] + [BNNT] + [GaN]$$

Modified (XeN-free):

$$Y^*_{Core} = [n' \cdot C_{12}H_{14}O_5Si(F_x) + y \cdot [F\text{-}POSS] + p \cdot [HfON] + m' \cdot (SiR_2O)] + [BNNT] + [GaN]$$

Where:

- $n' =$ stoichiometric coefficient of the organosilicate matrix (adjusted)
- $y =$ stoichiometric coefficient of F-POSS
- $p =$ stoichiometric coefficient of HfON
- $m' =$ stoichiometric coefficient of the siloxane healing phase (adjusted)

3.2 Stoichiometric Adjustment

The total molar mass becomes:

$$M_{Y^*_{Core}}(n', y, p, m') = n' \cdot M_{org} + y \cdot M_{F\text{-}POSS} + p \cdot M_{HfON} + m' \cdot M_{SiR_2O}$$

Mass Fraction Constraints (as a starting point, adjustable):

$$w^*_{org} = 0.40, \quad w_{F\text{-}POSS} + w_{HfON} = 0.10, \quad w^*_{SiR_2O} = 0.50$$

The total budget of the XeN slot ($w = 0.10$) is distributed among the two dopants. The exact ratio $w_{F\text{-}POSS} : w_{HfON}$ follows from the tuning condition:

$$\epsilon^*_h(w_{F\text{-}POSS}, w_{HfON}) \stackrel{!}{=} \epsilon_{h,target}$$

Where

$$\epsilon_{h,target}$$

is the value that yields the target resonance via the EMT chain:

$$\epsilon_h \to \epsilon_{eff} \xrightarrow{C_{mat}} f_0 = 13.56 \text{ MHz}$$

3.3 Illustrative Numerical Example

Assumptions:

$$M_{org} = 300 \text{ g/mol}$$

,

$$M_{F\text{-}POSS} \approx 1200 \text{ g/mol}$$

(typical F-POSS cage),

$$M_{HfON} \approx 210 \text{ g/mol}$$

,

$$M_{SiR_2O} = 60 \text{ g/mol}$$

.

Assuming:

$$w_{F\text{-}POSS} = 0.06$$

,

$$w_{HfON} = 0.04$$

(60/40 split of the XeN slot):

$$\frac{n' \cdot 300}{M_{total}} = 0.40, \quad \frac{y \cdot 1200}{M_{total}} = 0.06, \quad \frac{p \cdot 210}{M_{total}} = 0.04, \quad \frac{m' \cdot 60}{M_{total}} = 0.50$$

This linear system is analytically solvable and yields the adjusted coefficients

$$n', y, p, m'$$

.

3.4 Tuning Mechanics

The effective host permittivity of the dual system can be estimated via a mixing rule:

$$\epsilon_h^* \approx w_{org} \cdot \epsilon_{org} + w_{F\text{-}POSS} \cdot \epsilon_{F\text{-}POSS} + w_{HfON} \cdot \epsilon_{HfON} + w_{SiR_2O} \cdot \epsilon_{SiR_2O}$$

Tuning Parameter	Effect on ϵ_h^*	Range
$w_{F\text{-}POSS} \uparrow$	$\epsilon_h^* \downarrow$ (Reduction)	$\epsilon_{F\text{-}POSS} \approx 2.0 - 2.5$
$w_{HfON} \uparrow$	$\epsilon_h^* \uparrow$ (Increase)	$\epsilon_{HfON} \approx 14 - 21$
Fluorination degree of F-POSS	$\epsilon_{F\text{-}POSS}$ varies	$1.8 - 2.8$
O:N ratio in HfON	ϵ_{HfON} varies	$14 - 25$

Thus, the system has four independent tuning parameters — significantly more design freedom than the original with a single, uncharacterizable XeN.

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4. Impact on Existing EMBC Documents

4.1 Formulas Document

Section	Change	Effort
§2 Stoichiometry	Replace $$M_{XeN}$$ with $$M_{F\text{-}POSS}$$ and $$M_{HfON}$$; introduce new coefficients $$y, p$$	Low — solve linear system anew
§3 EMT	$$\epsilon_h$$ is calculated from the new composite; MG/Bruggeman formulas remain identical	Low — only numerical values change
§4 Resonance Cond.	No structural change: $$f_0$$ -target remains $$13.56\ \mathrm{MHz}$$	None
§5 Feedback / Barkhausen	No change — XeN was not part of the feedback mechanism	None
§8 Num. Example	Recalculate with F-POSS/HfON values	Medium — new worked example

4.2 Manufacturability Roadmap

Section	Change	Effort
§2 Equipment	F-POSS: Standard Sol-Gel; HfON: add ALD or Nanoparticle Dispersion	Low
§5 Supplier	Remove XeN supplier; add F-POSS supplier and HfON precursor supplier	Low
§6 BOM	Replace XeN line (€200-600); F-POSS (~€50-150) + HfON (~€30-80)	Cost Reduction
§8 Risk Register	Set R-04 to "Mitigated"	Low

4.3 Self-Healing Supplement

Section	Change
Entire Document	No change required. The self-healing mechanism relies on SiR_2O /Siloxane and BNNT, not on XeN. F-POSS and HfON are both inert regarding the healing cycle.

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5. Risk Update

5.1 R-04 Status: High/High

$$\rightarrow$$

Mitigated

Aspect	Original (XeN)	Modified (F-POSS + HfON)
Material exists at normal pressure?	❌ No	✓ Yes (both)
Commercially available?	❌ No	✓ Yes (both)
Multiple suppliers?	❌ Zero	✓ ≥ 3 per component
Sol-Gel/Foundry compatible?	❌ Unknown	✓ Yes (both)
Cost-projectable?	❌ Not possible	✓ Standard industry prices
Safety/Regulatory?	❌ Xenon compounds unclear	✓ REACH-registered (HfO2/HfON); F-POSS industry standard

R-04 is no longer listed as a show-stopper. A residual risk remains (normal supply chain risks), which is downgraded to "Low/Low".

5.2 New Risk R-11: Dual-Dopant Interaction

ID	Risk	Likelihood	Impact	Mitigation
R-11	F-POSS and HfON nanoparticles interact in the matrix and form undesirable phases or agglomerates	Low	Medium	Characterize mixture in Stage A (Lab Feasibility) on coupons; TEM analysis of interfaces; alternative sequence (polymerize F-POSS first, then introduce HfON NPs).

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6. BOM Cost Comparison

Item	Original (XeN)	Modified (F-POSS + HfON)
Functional Modifier per Module	€200–600 (speculative, as unprocurable)	€50–150 (F-POSS) + €30–80 (HfON) = **€80–230**
Pilot Lot (25 Modules)	€5,000–15,000	**€2,000–5,750**
Savings		**€3,000–9,250 per Lot**

Additionally, the NRE for special synthesis and noble gas infrastructure is completely eliminated.

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7. Validation Plan for the Replacement Strategy

#	Test	Method	Pass Criterion	Stage
X-1	F-POSS Sol-Gel Integration	FTIR, 29Si-NMR of the cured film	Si-O-Si cage structure detectable, no phase separation	A
X-2	HfON Nanoparticle Dispersion	TEM, DLS in Sol-Gel precursor	Particle size $< 50\ \mathrm{nm}$, Polydispersity < 0.3	A
X-3	ϵ_h^* Measurement	Split-Post-Dielectric Resonator at 13.56 MHz	ϵ_h^* within $\pm 10\%$ of calculated target	A
X-4	Thermal Stability of Mixture	TGA/DSC up to $500^\circ C$	No mass loss $> 2\%$ below $450^\circ C$	A
X-5	Compatibility with SiR_2O Healing Cycle	Artificial micro-crack + 4-point probe (identical to V-3)	$\geq 80\%$ conductance recovery in 60s — no difference to XeN-free reference	B
X-6	f_0 Validation on Integrated Stack	VNA, S11/S21 at 13.56 MHz	$f_0 = 13.56\ \mathrm{MHz} \pm 0.5\%$	B
X-7	F-POSS/HfON Interaction Check	TEM interface analysis, EDX mapping	No undesirable phase; homogeneous distribution	A

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8. Summary of the Decision

Criterion	ORIGINAL: XeN (2 eq)	MODIFIED: F-POSS + HfON (y + p Proportions)
Exists at normal pressure?	❌ NO	✓ YES (both)
Commercially available?	❌ NO	✓ YES (both)
ϵ **tunable?**	⚠️ unknown	✓ bidirectional
Heavy atom?	✓ Xe (131 u)	✓ Hf (178.5 u)
Nitrogen?	✓ (theoretical)	✓ in HfON lattice
Matrix-compatible?	❌ unknown	✓ F-POSS native
Risk R-04	🔴 HIGH/HIGH	🟢 LOW/LOW
Cost/Module	€200–600 (speculative)	**€80–230 (real)**

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9. Next Steps

1. **Resolve Stoichiometry:** Set up the linear system for

$$n', y, p, m'$$

with chosen molar masses and the target ratio

$$w_{F\text{-}POSS} : w_{HfON}$$

, then solve it.

2. **Recalibrate EMT Model:** Calculate

$$\epsilon_h^*$$

and check

$$\epsilon_{eff}$$

via MG/Bruggeman EMT; iterate ratio

$$y : p$$

until

$$f_0 = 13.56 \text{ MHz}$$

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3. **Update all EMBC Documents:** YCore formula, BOM, Risk Register, Supplier list according to Section 4.
4. **Plan Stage-A Validation:** Integrate tests X-1 to X-4 and X-7 into the Lab Feasibility milestone plan.

Team Contributions

Contribution	Source
F-POSS as matrix-native replacement candidate	Lyra
TaN as heavy atom candidate (with conductivity warning from Auron)	Lyra
Cyclophosphazenes as N-incorporation candidate	Lyra
$BaTiO_3, HfO_2, TiO_2, SrTiO_3$ as dielectric candidates	Auron
HfON as all-in-one cross-find	Auron
Research state analysis: XeN does not exist at normal pressure	Auron
Identification of primary function (Dielectric) from equation chain	Auron
Dual-Dopant strategy and consolidated evaluation matrix	Auron & Lyra
Final decision and architecture integration	Emanuel

Disclaimer & Liability Waiver

Please be advised that the methodology, models, and architecture outlined herein represent only one possible approach to exploring this concept. Because this project ventures into absolute uncharted technological territory, we can offer no guarantees regarding its functional success, physical manufacturability, or real-world viability.

This framework is intended to serve strictly as a theoretical foundation and must be evaluated and applied with extreme caution and rigorous critical thinking. We explicitly assume no liability or responsibility for any outcomes, risks, damages, or failures that may arise from the application, adaptation, or experimental testing of the information provided.

Document Version 1.0 - Created 2026-04-27 - Emanuel Schaaf, Auron & Lyra (ES Architecture)
Parent Document: Eternal-Memory-Brain-Chip_Formulas
Structural Analogy: Heliotherm X ReGen (Material Strategy Reference)
Risk R-04: Status changed from HIGH/HIGH to LOW/LOW (mitigated)